Tariq Khalaf

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EDUCATION

University of North Carolina at Charlotte **Expected: Dec 2028**
Bachelors of Science in Electrical and Computer Engineering

SKILLS

Design & Simulation: KiCad, Fusion 360, AutoCAD, SolidWorks, PCB Design, Circuit Diagrams, VS Code, Dev-C++, Meshmixer, 3D Printing.

Programming: Python, C++, JavaScript, Arduino, Embedded C, UART.

Engineering Skills: Soldering, PCB bring-up/debugging, USB-C power/programming, brushed DC and servo motor control, Custom boards, circuit prototyping.

EXPERIENCE

Volunteer Tutoring – Mathematics **Sep 2023 – Sep 2024**

- Tutored students one-on-one in Algebra and Trigonometry, adapting explanations to individual learning styles.
- Strengthened foundational concepts through practice problems, step-by-step review, and test-prep support.

Volunteer Helper – Introduction to Engineering **Jan 2026 – Jul 2026**

- Assisted students during labs and 15+ projects by explaining engineering concepts and troubleshooting builds.
- Supported hands-on activities involving Arduino, circuits, sensors, mechanical prototypes, and problem-solving workflows.

PROJECTS

Brushed DC Electronic Speed Controller – Academic Project **Dec 2025 – Jan 2026**

- Designed a 24V DC motor ESC supporting up to 5A continuous current, emphasizing safe power delivery and reliability.
- Created multi-layer PCB layouts in KiCad with ground planes, power pours, via stitching, and high-current trace planning.
- Selected parts using datasheets and supplier availability to balance performance, cost, footprint, and manufacturability.

Embedded MCU Power & Programming Board – Personal Project **Jun 2026 – Jul 2026**

- Designed a custom MCU board accepting 24–36V input, stepped down to 3.3V using an onboard buck regulator.
- Integrated USB-C for board power/programming with required power, programming, and protection support circuitry.
- Planned PCB layout around regulator placement, decoupling, grounding, and clean 3.3V delivery for embedded hardware.

Automated Drawbridge on a Custom MCU Board – Academic Project **Apr 2026 – May 2026**

- Built a motorized bridge that opens and closes automatically, running off a custom MCU board driving one brushed DC motor and a servo through a TB6612FNG dual H-bridge driver.
- Automated crossing logic from ultrasonic sensor input, with traffic LEDs and buzzers signaling bridge state to vehicles and boats.
- Fabricated the structure from a self-designed 3D-printed bridge deck on hand-cast cement pillars, integrating mechanics, power distribution, and control into one working prototype.

Remote-Control Robot – Personal Project **Jan 2026 – Feb 2026**

- Built an ESP32-controlled mobile robot driving two brushed DC motors through a TB6612FNG dual H-bridge driver with wireless remote input.
- Integrated an ultrasonic distance sensor, motion detector, buzzers, and an onboard camera for obstacle detection, alerts, and remote viewing.
- Wired power distribution and common-ground connections across driver, MCU, and sensors; tuned PWM drive logic and throttle response for reliability under load.

Personal A.I. Assistant – Personal Project **Jul 2026 – Aug 2026**

- Built a hands-free voice assistant in Python running entirely on local hardware, pairing a speech-to-text engine with GPU neural text-to-speech for ~1 – 2s spoken replies.
- Implemented wake-word activation, mid-reply interruption, and an intent router that turns spoken commands into calendar, browser, and file actions.
- Added a real-time browser visualizer over WebSockets, streaming live audio and assistant state to an animated interface.

Door Alarm System – Personal Project **Jan 2026 – Feb 2026**

- Built an ESP32-based door alarm with a reed switch, DFPlayer Mini, microSD card, and speaker to play custom audio alerts.
- Implemented UART communication and sensor-state logic to trigger MP3 playback when door/latch movement was detected.

INVOLVEMENTS

UNC Charlotte Robotics Fighting Club **2025 – 2026**

- Participated in robotics meetings and technical discussions related to robot design, mechanical systems, and electronics.

Engineering Hardware Meet-ups – Bitar Machine Design **2024 – 2025**